

YAROSLAV PRYTULA

COMPUTER VISION RESEARCHER

I am a first-year PhD student in Computer Science at the University of Tartu, focusing on machine learning and computer vision. I work as a Research Engineer at the Biomedical Computer Vision Lab on transformer-based segmentation and detection in medical imaging.

EDUCATION

MATHEMATICS AND COMPUTER SCIENCE

PhD, Ukrainian Catholic University and University of Tartu

(09/2025 - present)

COMPUTER SCIENCE

MSc, University of Tartu

(09/2023 - 06/2025)

COMPUTER SCIENCE

BSc, Ukrainian Catholic University

(08/2019 - 06/2023)

PHYSICS AND MATHEMATICS

Lviv Physics and Mathematics Lyceum

(08/2015 - 05/2019)

PUBLICATIONS

- **Yaroslav Prytula**, Illia Tsiporenko, Ali Zeynalli, and Dmytro Fishman, "IAUNet: Instance-Aware U-Net," accepted for presentation at the **CVMI Workshop, CVPR 2025**.

TALKS & LECTURES

Computer Vision in Agriculture

Delivered as part of Data Science Summer School, Ukrainian Catholic University, summer 2025.

Multimodal Generative Models in Computer Vision

Delivered as part of the GenAI Spring School, Ukrainian Catholic University, 2025.

Covered diffusion-based text-to-image generation models including DALL-E, Stable Diffusion, and FLUX. Focused on architecture and core concepts.

Zero-Shot Models in Computer Vision

Ukrainian Catholic University, 2025.

Lecture on zero-shot models in computer vision, starting with Vision Transformers and covering CLIP, SAM, FastSAM, and SAM2, Grounding DINO, and Grounded SAM for tracking.

s.prytula@ucu.edu.ua

yaroslav.prytula@ut.ee

Ukraine, Lviv

linkedin.com/in/yaroslav-prytula-3166471b0

github.com/SlavkoPrytula

kaggle.com/slavkoprytula

WORK EXPERIENCE

Computer Vision Researcher

BCVL Research Group, Tartu

(03/2022 - present)

I am currently working on a few research projects in the field of medical computer vision under the guidance of my Ph.D. supervisor, Dmytro Fishman. We are working on state-of-the-art foundational models for medical imaging. Currently focusing on segmentation tasks under heavy occlusions and high-density instance segmentation in microscopy using transformer models and self-supervised learning.

ML Lecturer

Ukrainian Catholic University

(09/2024 - present)

I teach the Machine Learning module of the Artificial Intelligence course and supervise student research projects focused on AI, Machine Learning, and Computer Vision.

Co-Instructor

University of Tartu

(02/2025 - present)

Co-instructing the Master's course Deep Learning for Computer Vision at the University of Tartu. Gave lectures on Vision Transformers for classification, segmentation, and object detection.

Data Scientist

Ukrainian Catholic University ML Lab

(09/2021 - 05/2024)

In collaboration with SoftServe's R&D department, we develop video-based real-time vital signs capture project. The task requires the analysis of frequency domains of the extracted signal from the region of interest on the face. We have developed a strong algorithm for detecting and tracking heart rate from real-time videos. We are now exploring unsupervised computer vision and signal processing approaches for more robustness in different environmental conditions.

RESEARCH PROJECTS

HuBMAP + HPA - Hacking the Human Body

(07/2022 - 10/2022)

Kaggle competition on identifying and segmenting functional tissue units (FTUs) across five human organs
(26 place) (top 3%)

Sartorius - Cell Instance Segmentation

(09/2021 - 01/2022)

Kaggle competition on detection and segmentation of neuronal cell instances in microscopy images
(143 place) (top 10%)

TensorFlow - Help Protect the Great Barrier Reef

(11/2021 - 02/2022)

Kaggle competition on crown-of-thorns starfish detection and tracking underwater (26 place) (top 1%)

RSNA-MICCAI Brain Tumor Radiogenomic Classification

(07/2021 - 09/2021)

Kaggle competition on predicting a probability for the target of a malignant tumor in the brain (glioblastoma)

SKILLS

Python, C++, Java, C, PyTorch, PyTorch Lightning, TorchScript (familiar), ONNX (familiar), TensorRT (familiar), MMDetection, Detectron2, OpenCV, real-time inference, model quantization (familiar), Git, Docker, gRPC, CMake, SLURM, Weights & Biases, AWS (familiar), GCP (familiar), research-to-production workflows, team collaboration, time management, research and publication experience

INTERESTS

Medical Imaging, Computer Vision, Machine Learning, Self-Supervised Learning, Transformers, Signal Processing

SCHOLARSHIPS / AWARDS

- Second Best MSc Thesis Award, University of Tartu (2025)
- Research Stipend (2025)
- Tuition Waiver (2025)
- Achievement Stipend (2024)

CERTIFICATES

Oxford Machine Learning School

(05/2025 - 08/2025)

OxML@2025. Oxford Machine Learning School. Oxford's Deep Medicine & CIFAR. MLx Generative AI. Representation Learning & Gen.AI

Lviv Data Science Summer School

(08/2024) - Ukrainian Catholic University

Lviv Data Science Summer School (LDSSS@2024)

Improving Deep Neural Networks

(07/2021 - 08/2021)

Coursera course on Deep Learning

IELTS

(03/2023)

7.5 Overall Band Score (C1 CEFR Level)

LANGUAGES

English

Full Professional Proficiency (C1)

Ukrainian

Native or Bilingual Proficiency

VOLUNTEERING

Open-Door Event for Entrants

(04/2021)

Hosted an open-door event for entrants with my peers for our university faculty. I have communicated with a team that would help younger students understand more about the Linux club on our faculty, what we do and how to apply there

Summer Data Science School

(07/2021)

Participated as a volunteer for a two-week Summer School of Data Science that gathered participants from around the world to give practice-oriented knowledge about areas of it

Math Olympiad

(02/2021)

Participated as one of the organizers of the Math Olympiad for middle school classes